DO NOW — RECALL

Lesson 082 · Quicksort: Partition Around a Pivot

Name: _____ Class: _____ Date: _____

1. What is merge sort's combine step?

2. What is a pivot in quicksort?

3. Is a partitioned array necessarily fully sorted?

4. Swap values at indexes0 and2 in [4,1,7]. Give the resulting array.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Merge two individually sorted runs.

2. Model answer

Selected value used to partition an array.

3. Model answer

No; subarrays still need sorting.

4. Model answer

[7,1,4].

By the end, you can…

• Partitioning reorders values around a chosen pivot so values to its left are no larger and those right are larger, for this partition rule.

• The pivot's final index is fixed after partition, but neither side is necessarily fully sorted; quicksort continues with the two subarrays.

• Average and best-case time O(n log n), worst O(n^2) with poor pivot choices; recursion consumes stack space depending on partition balance.

Vocabulary — know these terms

quicksort — Recursive partition-based sorting.

pivot — Value partitioning subarray.

partition — Place smaller/larger values around pivot.

worst case — Maximum growth for adverse input.

NEW LEARNING

Read, annotate and explain

Observe

Partitioning reorders values around a chosen pivot so values to its left are no larger and those right are larger, for this partition rule.

Apply

The pivot's final index is fixed after partition, but neither side is necessarily fully sorted; quicksort continues with the two subarrays.

Evaluate

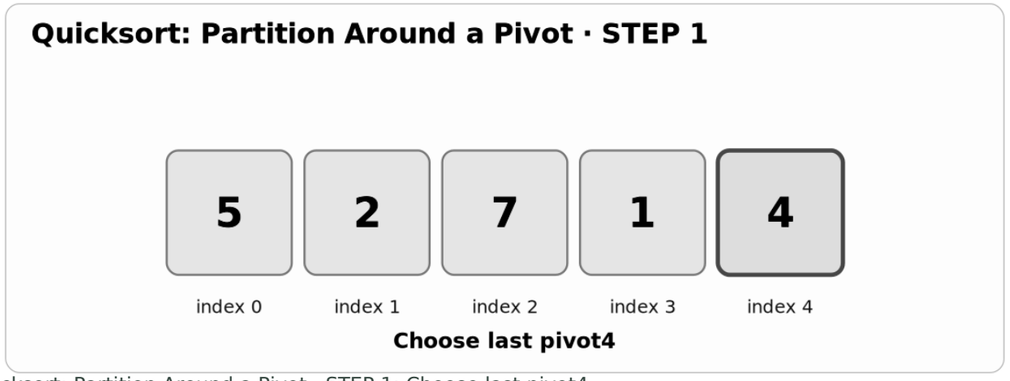
Average and best-case time O(n log n), worst O(n²) with poor pivot choices; recursion consumes stack space depending on partition balance.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

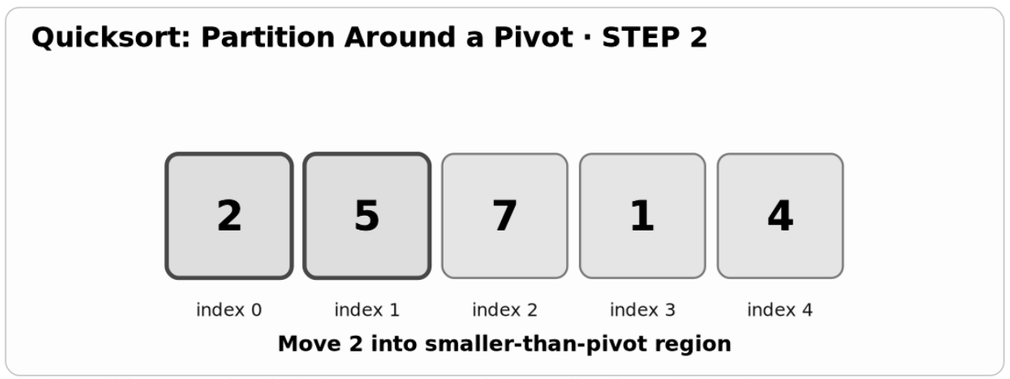
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Quicksort: Partition Around a Pivot · STEP 1; Choose last pivot4

New Learning



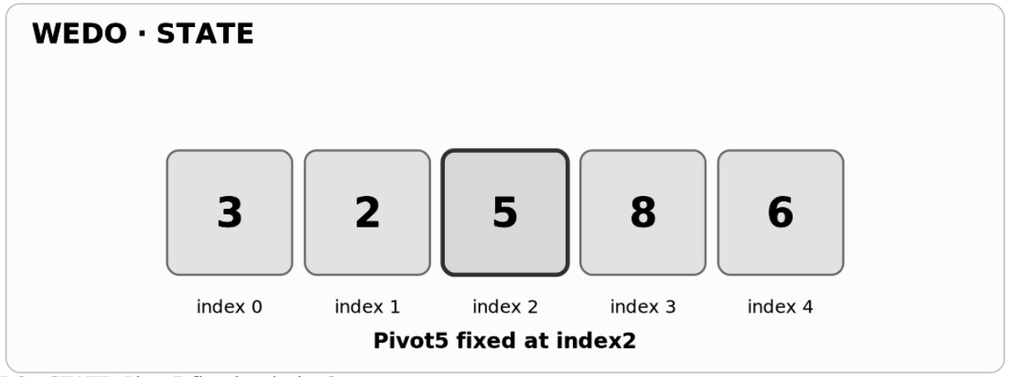
Quicksort: Partition Around a Pivot · STEP 2; Move 2 into smaller-than-pivot region

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Partition [8,3,6,2,5] using last pivot5.

Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].



WEDO · STATE; Pivot5 fixed at index2

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

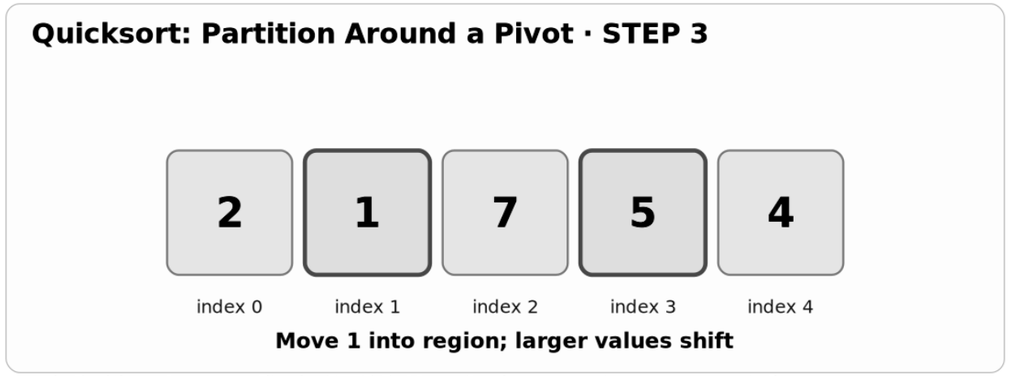
Guided 1 — CHECK

Method and result
Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].

New Learning · original lesson diagrams

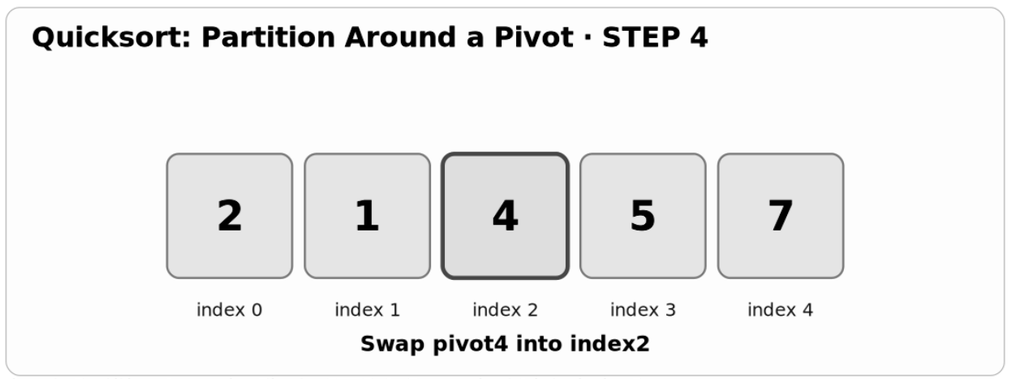
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Quicksort: Partition Around a Pivot · STEP 3; Move 1 into region; larger values shift

New Learning



Quicksort: Partition Around a Pivot · STEP 4; Swap pivot4 into index2

I DO — WORKED EXAMPLE

Follow the worked method

Partition [6,2,5,1,3] using last pivot3; give partition state.

Method and result

Lomuto partition moves 2 and 1 left, pivot3 to index2 → [2,1,3,6,5].

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Once a pivot reaches its correct final index, both partitions are also completely sorted.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

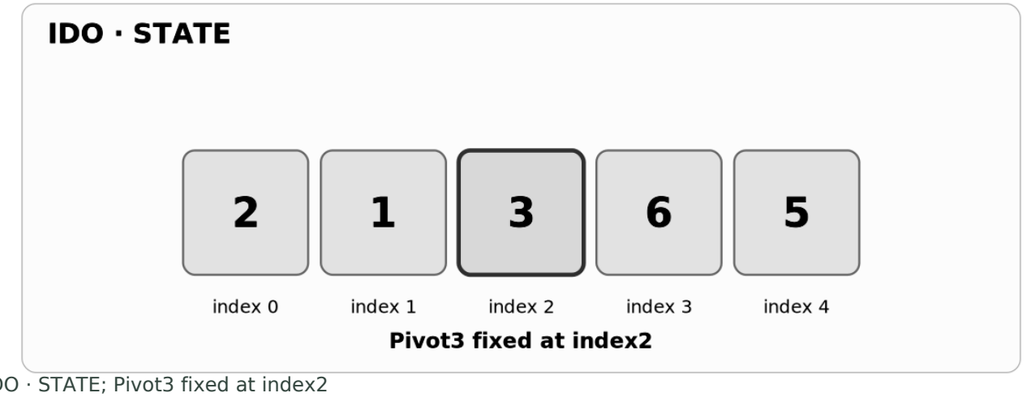
Explain how pivot choice impacts quicksort worst-case time and recursive stack depth. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Partition [6,2,5,1,3] using last pivot3; give partition state.

Lomuto partition moves 2 and 1 left, pivot3 to index2 → [2,1,3,6,5].



IDO · STATE; Pivot3 fixed at index2

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Partition [8,3,6,2,5] using last pivot5.
Source code / input:
Smaller 3,2 move left; pivot5 at index2 → [3,2,5,8,6].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace Lomuto pivot-last partition of [5,2,7,1,4] and give final pivot index. [6]

Question 2

After quicksort partition gives [2,1,4,5,7] with pivot4 at index2, state required recursive subranges and eventual sorted result. [6]